HS-RWE Data Challenge

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1. Problem and approach

We forecast each patient's average IHS4 for January–June 2026 from records ending 31 December 2025 (21,000 train patients). The records only see the disease on visit days: 36% of 709,690 visits are unscored, and scores cluster around flares (12.8 at flare visits vs 9.0 at background visits). We treated these gaps as signal, not noise.

2. Features

One function builds one row per patient, identically for train, test_1 and test_2, from data up to 31 December 2025 only.

- **Smoothed IHS4:** exponentially weighted means (half-life 90 and 365 days), plus a "calm level" from non-flare visits only.
- **Observation pattern:** visits by type, share scored, days since last visit and last flare, longest gap, flare rate and its recent change.
- **Treatment and history:** current biologic, drug, days on it, failed lines, antibiotics, surgeries, Hurley stage, missing-value flags.

Seasonality was flat (21–22% flare visits in every month), so calendar features were not needed.

3. Results

Exercise	Patients	Result
1. Forecast	21,000 train, 5-fold CV	XGBoost MAE 1.47 vs 2.65 for the best baseline (smoothed IHS4): 45% lower
2a. Biologic forecast	8,548 on a biologic	Dedicated model MAE 1.21 vs 1.25 with the exercise 1 model
2b. Super-responders	6,793 labelled, 12.0% positive	ROC-AUC 0.74 before treatment, 0.80 after 6 months on the drug
3a. Images	2,800	MAE gain +0.03 overall, +0.14 for 661 recent biologic starters
3b. RNAseq	600 with both embeddings	MAE gain +0.38 [95% CI +0.29, +0.48]

Exercise 1. LightGBM, Random Forest and an ensemble all score 1.46–1.53, so the gain comes from the features, not the algorithm. Error is highest for recent biologic starters (2.24 within 3 months vs 1.03 after 2 years).

Super-responders (2026 IHS4 below 25% of the 12-month pre-biologic mean) are more often on adalimumab (14.9% vs 2–3% on IL-17 drugs, partly selection), on their first biologic (0.06 vs 0.32 earlier lines), earlier in the disease (fewer surgeries, lower Hurley stage), and respond fast (56% vs 83% of the pre-drug level after 6 months). Sex, smoking and ethnicity barely matter.

Embeddings. RNAseq, sampled when the biologic starts, captures the response the records cannot yet show; its gain vanishes after about 9 months on the drug. Images mostly re-encode the visit's flare severity, so 3a corrects only recent starters. Device, batch or date alone predict nothing.

4. Clinical signal vs recording artefacts

- **Genuine:** recent smoothed IHS4 (37% of the model's gain), calm level, Hurley stage, biologic treatment. The crude biologic association is only −0.37 points because the sickest patients are treated; adjusting for Hurley stage and baseline severity gives −4.23 [−4.38, −4.08].
- **Artefacts:** documentation features (share scored, gaps) reflect how sites record data. Adding the practitioner ID worsened MAE to 1.70. Missing BMI or smoking status carries no information.

5. test_1 vs test_2, and what did not work

test_1 is indistinguishable from train (adversarial AUC 0.50); test_2 is not (0.68). It has more Black (59% vs 27%) and female (89% vs 72%) patients, and only 43% of visits scored vs 60%. Expected test_2 MAE is about 1.60 vs 1.47, driven mainly by the thinner documentation. This check also exposed, and we fixed, a per-folder bug in the practitioner patient count.

Kalman-filtered levels, visit-type-adjusted scores, interpolation of unscored visits, trajectory clusters and practitioner encoding each changed MAE by less than 0.005. Limitations: one random-fold CV scheme, a rough super-responder proxy, and test_2 performance estimated rather than measured.